

FormFix-AI

MULTI-MODEL AI REAL TIME GYM POSE DETECTION AND CORRECTION USING
LSTM, BiLSTM and TRANSFORMER

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Mini Project Presentation

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Problem Statement

Many users perform workouts without supervision, which leads to poor exercise form, wrong repetition counting, low training quality, and injury risk. Existing systems usually solve only one piece of the problem, such as pose detection or action recognition, but do not combine recognition, repetition counting, posture correction, reporting, and user progress tracking into one deployable product.

Project goal

- detect the exercise being performed,
- count reps accurately,
- check posture in real time,
- generate voice and text feedback,
- save progress and end-of-exercise summaries.

What Was Implemented from Research Papers

- From MediaPipe/OpenPose papers: real-time body landmark extraction.
- From temporal learning papers: LSTM, BiLSTM, and Transformer sequence classification.
- From posture correction papers: joint-angle based biomechanical rules.
- From feedback-oriented papers: immediate guidance during workout performance.

What is added beyond the papers

- exercise-wise rep counting,
- friendly coach-style correction,
- live web camera interface,
- database-backed user progress tracking,
- session reports with perfect reps, corrected reps, and common mistakes.

Research Paper Comparison

Paper	Core idea	Real-time	Reported accuracy
Khadtare et al. (IJECE 2025)	MediaPipe + angle-based posture correction	Yes	Not clearly stated in extracted section
Gadhvi et al. (PosePilot)	LSTM/BiLSTM + attention for posture recognition/correction	Yes	97.52%
Appiah (VISAPP 2024)	mobile-phone pose estimation for gym correction	Near real-time	values in 90–97% range appear in extracted section
Kanase et al. (ICACC 2021)	OpenPose + vector geometry correction	Yes	no single final accuracy clearly stated
Suryawanshi et al. (IJRSI 2025)	OpenCV + MediaPipe + real-time corrective feedback	Yes	multiple percentages extracted; exact headline metric should be confirmed from full

Dataset Used

The project is trained on processed pose-sequence data. Each training sample is a **30-frame sequence**, and each frame contains **132 features** derived from detected body landmarks.

Processed dataset summary

- total sequences: **1140**
- sequence shape: (1140, 30, 132)
- classes: **6**

Class distribution

- barbell_squat: 152
- bicep_curl: 225
- hammer_curl: 78
- lateral_raise: 170
- pull_up: 212
- push_up: 303

Video Count vs Sequence Count

The model is trained on pose sequences, not directly on raw videos. One exercise video can generate multiple valid 30-frame windows, so the number of processed sequences is usually greater than the number of source videos.

Example

- hammer_curl videos added by user: 19
- processed hammer_curl sequences in dataset: 78

This is expected because one video may produce several usable training windows after pre-processing.

End-to-End Architecture



Preprocessing, Training and Evaluation Pipeline

Preprocessing

- 1 collect exercise videos,
- 2 extract pose landmarks using MediaPipe,
- 3 flatten frame-wise landmark features,
- 4 create 30-frame windows,
- 5 store processed dataset in `exercise_sequences.npz`.

Training

- 1 split data using stratified train/validation partition,
- 2 compute class weights,
- 3 train LSTM, BiLSTM, and Transformer,
- 4 save models in `models/`.

Evaluation

- 1 load trained model,

2 predict all sequences

Joint Angles Used for Rep Counting and Correction

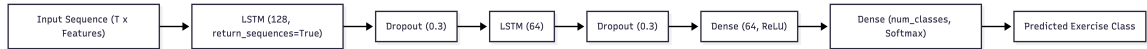
- Elbow angle = shoulder–elbow–wrist
- Knee angle = hip–knee–ankle
- Hip angle = shoulder–hip–knee
- Shoulder angle = elbow–shoulder–hip

$$\theta = \cos^{-1} \left(\frac{(A - B) \cdot (C - B)}{\|A - B\| \|C - B\|} \right)$$

These angles are reused by:

- rep counter,
- posture correction engine,
- report generator.

Baseline LSTM Architecture



Function: baseline temporal classifier.

Why used: simple and fast sequence-learning baseline for comparison.

BiLSTM Architecture



Function: main bidirectional temporal classifier.

Why used: captures motion context from both directions and improves exercise understanding.

Transformer Architecture



Function: attention-based sequence model.

Why used: captures long-range frame relationships and acts as an advanced comparison model.

How the Three Models Are Used

- LSTM, BiLSTM, and Transformer are trained on the same processed dataset.
- They are compared independently to identify the strongest architecture.
- The live system also supports hybrid inference:
 - fast mode favors LSTM,
 - best mode favors BiLSTM,
 - hybrid mode fuses all three with BiLSTM weight highest,
 - ensemble mode runs all models together.

Reason

- LSTM gives speed,
- BiLSTM gives strong temporal context,
- Transformer gives a high-capacity comparison model.

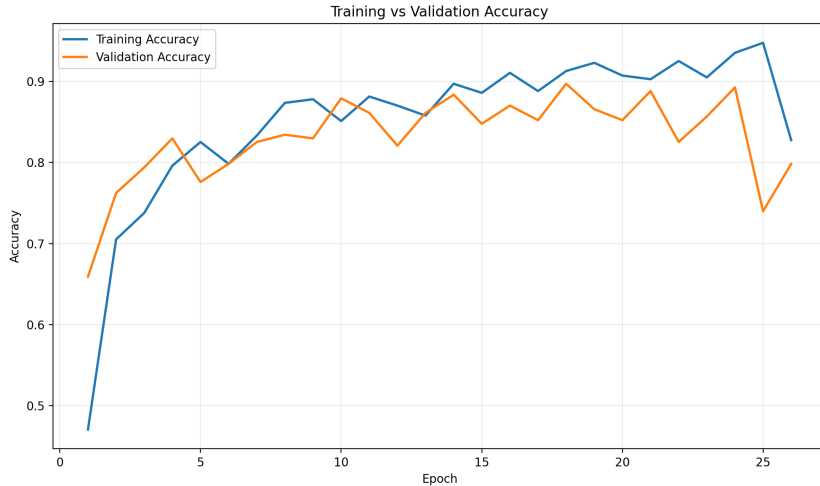
Training Log Snapshot: LSTM / BiLSTM

```
Epoch 1/22
187/187 [=====] - 8s 26ms/step - loss: 1.0406 - accuracy: 0.6286 - val_loss: 0.8880 - val_accuracy: 0.6622 - lr: 0.0010
Epoch 2/22
187/187 [=====] - 4s 20ms/step - loss: 0.8195 - accuracy: 0.7300 - val_loss: 0.7413 - val_accuracy: 0.7138 - lr: 0.0010
Epoch 3/22
187/187 [=====] - 4s 19ms/step - loss: 0.6887 - accuracy: 0.7722 - val_loss: 0.4806 - val_accuracy: 0.8157 - lr: 0.0010
Epoch 4/22
187/187 [=====] - 4s 19ms/step - loss: 0.5918 - accuracy: 0.8104 - val_loss: 0.5234 - val_accuracy: 0.8277 - lr: 0.0010
Epoch 5/22
187/187 [=====] - 4s 20ms/step - loss: 0.5860 - accuracy: 0.8364 - val_loss: 0.3967 - val_accuracy: 0.8485 - lr: 0.0010
Epoch 6/22
187/187 [=====] - 4s 19ms/step - loss: 0.4726 - accuracy: 0.8486 - val_loss: 0.2807 - val_accuracy: 0.8914 - lr: 0.0010
Epoch 7/22
187/187 [=====] - 4s 20ms/step - loss: 0.4136 - accuracy: 0.8667 - val_loss: 0.3230 - val_accuracy: 0.8881 - lr: 0.0010
Epoch 8/22
187/187 [=====] - 4s 20ms/step - loss: 0.3976 - accuracy: 0.8689 - val_loss: 0.2663 - val_accuracy: 0.9075 - lr: 0.0010
Epoch 9/22
187/187 [=====] - 4s 21ms/step - loss: 0.3591 - accuracy: 0.8840 - val_loss: 0.2862 - val_accuracy: 0.8914 - lr: 0.0010
Epoch 10/22
187/187 [=====] - 4s 20ms/step - loss: 0.3404 - accuracy: 0.8904 - val_loss: 0.2402 - val_accuracy: 0.9082 - lr: 0.0010
Epoch 11/22
187/187 [=====] - 4s 20ms/step - loss: 0.3089 - accuracy: 0.9016 - val_loss: 0.2610 - val_accuracy: 0.8928 - lr: 0.0010
Epoch 12/22
187/187 [=====] - 4s 19ms/step - loss: 0.3066 - accuracy: 0.8983 - val_loss: 0.2409 - val_accuracy: 0.9075 - lr: 0.0010
Epoch 13/22
186/187 [=====>.] - ETA: 0s - loss: 0.3257 - accuracy: 0.8987
Epoch 13: ReduceLROnPlateau reducing learning rate to 0.0005000000237487257.
187/187 [=====] - 4s 19ms/step - loss: 0.3252 - accuracy: 0.8989 - val_loss: 0.2502 - val_accuracy: 0.9048 - lr: 0.0010
Epoch 14/22
187/187 [=====] - 4s 20ms/step - loss: 0.2174 - accuracy: 0.9279 - val_loss: 0.1766 - val_accuracy: 0.9236 - lr: 5.0000e-04
Epoch 15/22
187/187 [=====] - 4s 19ms/step - loss: 0.1882 - accuracy: 0.9351 - val_loss: 0.1605 - val_accuracy: 0.9430 - lr: 5.0000e-04
Epoch 16/22
187/187 [=====] - 4s 20ms/step - loss: 0.1877 - accuracy: 0.9403 - val_loss: 0.1279 - val_accuracy: 0.9511 - lr: 5.0000e-04
Epoch 17/22
187/187 [=====] - 4s 19ms/step - loss: 0.1829 - accuracy: 0.9410 - val_loss: 0.1472 - val_accuracy: 0.9424 - lr: 5.0000e-04
Epoch 18/22
187/187 [=====] - 4s 19ms/step - loss: 0.1553 - accuracy: 0.9495 - val_loss: 0.1248 - val_accuracy: 0.9538 - lr: 5.0000e-04
Epoch 19/22
187/187 [=====] - 4s 19ms/step - loss: 0.1512 - accuracy: 0.9506 - val_loss: 0.1172 - val_accuracy: 0.9564 - lr: 5.0000e-04
Epoch 20/22
187/187 [=====] - 5s 28ms/step - loss: 0.1429 - accuracy: 0.9531 - val_loss: 0.1392 - val_accuracy: 0.9491 - lr: 5.0000e-04
Epoch 21/22
187/187 [=====] - 4s 20ms/step - loss: 0.1456 - accuracy: 0.9499 - val_loss: 0.1211 - val_accuracy: 0.9611 - lr: 5.0000e-04
Epoch 22/22
185/187 [=====>.] - ETA: 0s - loss: 0.1680 - accuracy: 0.9454
Epoch 22: ReduceLROnPlateau reducing learning rate to 0.0002500000118743628.
187/187 [=====] - 4s 20ms/step - loss: 0.1677 - accuracy: 0.9454 - val_loss: 0.1199 - val_accuracy: 0.9538 - lr: 5.0000e-04
2026-04-13 00:14:34,317 | train_lstm | INFO | Saved BiLSTM model to models/exercise_bilstm.keras
2026-04-13 00:14:34,319 | train_lstm | INFO | Saved LSTM model to models/exercise_lstm.keras
2026-04-13 00:14:34,320 | train_lstm | INFO | Final validation accuracy: 0.9799
```

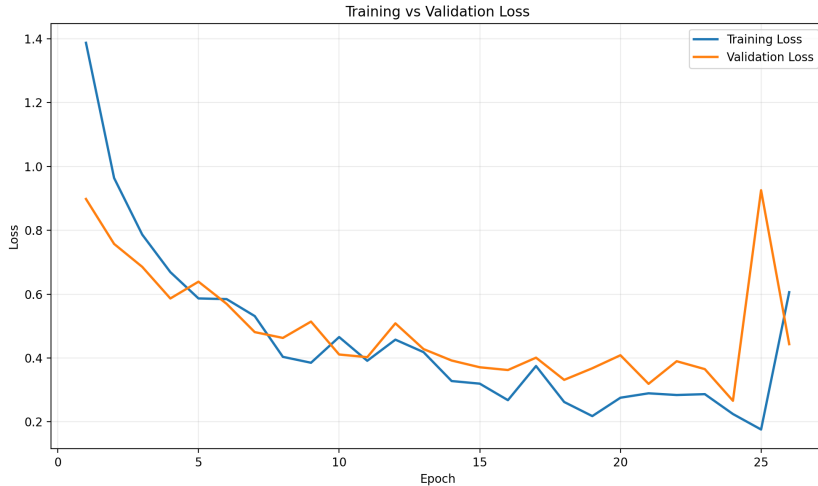
Training Log Snapshot: Transformer

```
Epoch 1/30
2026-04-13 00:14:44.482117: I tensorflow/core/grappler/optimizers/custom_graph_optimizer_registry.cc:117] Plugin optimizer for device_type GPU is enabled.
187/187 [=====] - 8s 36ms/step - loss: 0.0476 - accuracy: 0.6743 - val_loss: 0.5449 - val_accuracy: 0.7830
Epoch 2/30
187/187 [=====] - 6s 30ms/step - loss: 0.4489 - accuracy: 0.8364 - val_loss: 0.3719 - val_accuracy: 0.8572
Epoch 3/30
187/187 [=====] - 6s 30ms/step - loss: 0.3618 - accuracy: 0.8699 - val_loss: 0.3181 - val_accuracy: 0.8921
Epoch 4/30
187/187 [=====] - 8s 41ms/step - loss: 0.2045 - accuracy: 0.8964 - val_loss: 0.3333 - val_accuracy: 0.8854
Epoch 5/30
187/187 [=====] - 13s 71ms/step - loss: 0.2688 - accuracy: 0.9084 - val_loss: 0.2444 - val_accuracy: 0.9289
Epoch 6/30
187/187 [=====] - 7s 36ms/step - loss: 0.2148 - accuracy: 0.9179 - val_loss: 0.1675 - val_accuracy: 0.9358
Epoch 7/30
187/187 [=====] - 7s 39ms/step - loss: 0.1892 - accuracy: 0.9298 - val_loss: 0.1714 - val_accuracy: 0.9471
Epoch 8/30
187/187 [=====] - 7s 37ms/step - loss: 0.1999 - accuracy: 0.9211 - val_loss: 0.2446 - val_accuracy: 0.9848
Epoch 9/30
187/187 [=====] - 7s 35ms/step - loss: 0.1762 - accuracy: 0.9316 - val_loss: 0.1754 - val_accuracy: 0.9438
Epoch 10/30
187/187 [=====] - 8s 45ms/step - loss: 0.1658 - accuracy: 0.9373 - val_loss: 0.1874 - val_accuracy: 0.9438
Epoch 11/30
187/187 [=====] - 7s 37ms/step - loss: 0.1688 - accuracy: 0.9383 - val_loss: 0.1861 - val_accuracy: 0.9383
Epoch 12/30
187/187 [=====] - 7s 39ms/step - loss: 0.1458 - accuracy: 0.9468 - val_loss: 0.1884 - val_accuracy: 0.9417
Epoch 13/30
187/187 [=====] - 7s 35ms/step - loss: 0.1187 - accuracy: 0.9541 - val_loss: 0.1449 - val_accuracy: 0.9568
Epoch 14/30
187/187 [=====] - 6s 33ms/step - loss: 0.1887 - accuracy: 0.9588 - val_loss: 0.1647 - val_accuracy: 0.9497
Epoch 15/30
187/187 [=====] - 7s 35ms/step - loss: 0.1328 - accuracy: 0.9469 - val_loss: 0.1492 - val_accuracy: 0.9497
Epoch 16/30
187/187 [=====] - 6s 34ms/step - loss: 0.1117 - accuracy: 0.9568 - val_loss: 0.1562 - val_accuracy: 0.9497
Epoch 17/30
187/187 [=====] - 7s 36ms/step - loss: 0.1864 - accuracy: 0.9688 - val_loss: 0.1378 - val_accuracy: 0.9578
Epoch 18/30
187/187 [=====] - 6s 33ms/step - loss: 0.0923 - accuracy: 0.9646 - val_loss: 0.1245 - val_accuracy: 0.9645
Epoch 19/30
187/187 [=====] - 7s 36ms/step - loss: 0.1082 - accuracy: 0.9581 - val_loss: 0.1489 - val_accuracy: 0.9491
Epoch 20/30
187/187 [=====] - 6s 35ms/step - loss: 0.1180 - accuracy: 0.9584 - val_loss: 0.1598 - val_accuracy: 0.9551
Epoch 21/30
187/187 [=====] - 8s 44ms/step - loss: 0.0912 - accuracy: 0.9661 - val_loss: 0.1786 - val_accuracy: 0.9418
Epoch 22/30
187/187 [=====] - 6s 32ms/step - loss: 0.0908 - accuracy: 0.9661 - val_loss: 0.1256 - val_accuracy: 0.9658
Epoch 23/30
187/187 [=====] - 6s 34ms/step - loss: 0.0817 - accuracy: 0.9698 - val_loss: 0.1816 - val_accuracy: 0.9725
Epoch 24/30
187/187 [=====] - 6s 33ms/step - loss: 0.0781 - accuracy: 0.9732 - val_loss: 0.1421 - val_accuracy: 0.9611
Epoch 25/30
187/187 [=====] - 7s 39ms/step - loss: 0.0691 - accuracy: 0.9779 - val_loss: 0.1351 - val_accuracy: 0.9638
Epoch 26/30
187/187 [=====] - 18s 52ms/step - loss: 0.0909 - accuracy: 0.9626 - val_loss: 0.1683 - val_accuracy: 0.9524
Epoch 27/30
187/187 [=====] - 7s 40ms/step - loss: 0.0898 - accuracy: 0.9678 - val_loss: 0.1596 - val_accuracy: 0.9538
Epoch 28/30
187/187 [=====] - 9s 58ms/step - loss: 0.1829 - accuracy: 0.9614 - val_loss: 0.1368 - val_accuracy: 0.9638
Epoch 29/30
187/187 [=====] - 6s 34ms/step - loss: 0.1128 - accuracy: 0.9679 - val_loss: 0.1358 - val_accuracy: 0.9564
Epoch 30/30
187/187 [=====] - 6s 32ms/step - loss: 0.0688 - accuracy: 0.9787 - val_loss: 0.1185 - val_accuracy: 0.9685
2026-04-13 00:18:18.084 | train_transformer | INFO | Saved transformer model to models/posture_transformer.keras
2026-04-13 00:18:18.084 | train_transformer | INFO | Final validation accuracy: 0.9685
```

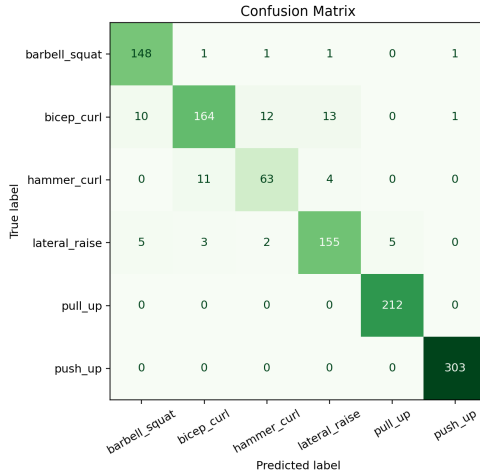
Training vs Validation Accuracy



Training vs Validation Loss



Confusion Matrix



How Accuracy is Calculated

Top-1 Accuracy

$$\text{Top-1} = \frac{\text{Correct highest-confidence predictions}}{\text{Total predictions}} \times 100$$

Top-3 Accuracy

$$\text{Top-3} = \frac{\text{True class inside top 3 predictions}}{\text{Total predictions}} \times 100$$

Precision, Recall, F1

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \quad F1 = 2 \cdot \frac{PR}{P + R}$$

From metrics_summary.json

- Top-1 Accuracy: **93.72%**
- Top-3 Accuracy: **99.82%**
- Weighted F1-score: **93.64%**
- Macro F1-score: **91.69%**

From bilstm_history.json

- training accuracy rises from $\approx 47\%$ to $\approx 94.7\%$
- validation accuracy peaks around **89.69%**
- late-epoch spikes in validation loss indicate overfitting risk

Three-Model Accuracy Comparison

Model	Top-1	Top-3	Comment
LSTM	84.30%	98.95%	stable baseline and faster model
BiLSTM (analytics export)	93.72%	99.82%	strong balance of accuracy and temporal learning
Transformer	97.89%	99.65%	strongest current evaluation result

Important note

- Transformer is strongest in the current evaluation run.
- For strict academic reporting, it should still be verified on a fully unseen holdout split before making the final deployment claim.

BiLSTM Class-wise Performance

Class	Precision	Recall	F1-score
barbell_squat	0.908	0.974	0.940
bicep_curl	0.916	0.820	0.865
hammer_curl	0.808	0.808	0.808
lateral_raise	0.896	0.912	0.904
pull_up	0.977	1.000	0.988
push_up	0.993	1.000	0.997

Interpretation

- strongest classes are push_up and pull_up,
- squat is strong,
- the most difficult class pair remains bicep_curl vs hammer_curl.

How This Project Improves on the Papers

- Many papers stop at posture detection or recognition only.
- This project combines:
 - exercise recognition,
 - repetition counting,
 - posture correction,
 - live voice feedback,
 - website deployment,
 - SQL-based progress tracking,
 - end-of-exercise reports.
- So the project is more complete as an end-to-end application, not only a model experiment.

Future Improvements

- add more balanced curl and squat videos,
- create a strict train/validation/test benchmark split,
- increase exercise coverage,
- add per-user calibration,
- improve exercise-quality scoring for each rep,
- optimize browser-side speed further for smoother live inference.

Conclusion

FormFix AI is a real-time AI workout assistant that combines pose estimation, temporal sequence modeling, rep counting, posture correction, web deployment, and user progress tracking.

The project uses LSTM, BiLSTM, and Transformer models for exercise classification, and complements them with exercise-specific angle logic for rep counting and corrective feedback. The current analytics show strong performance, especially for push-up, pull-up, and squat classes.

The system can be further improved through better dataset balance, stricter benchmarking, and expansion to more exercises, but it already demonstrates a practical AI fitness trainer built from research ideas and deployed as a usable application.

Thank You